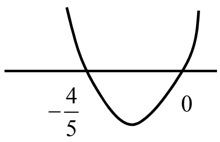
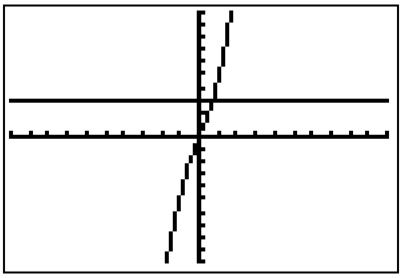
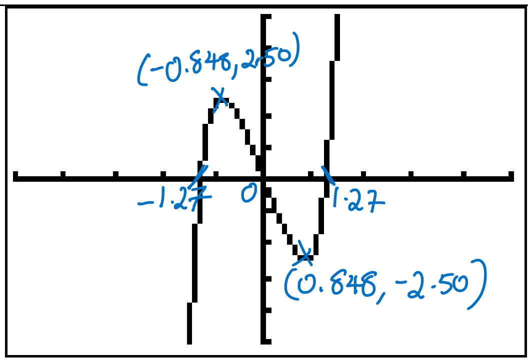
IJC Prelim 2 H1 Math Solutions

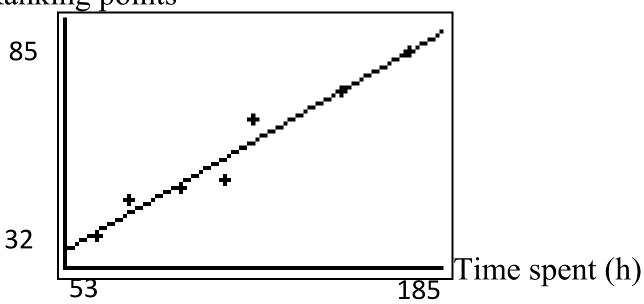
Section A (Pure Mathematics)

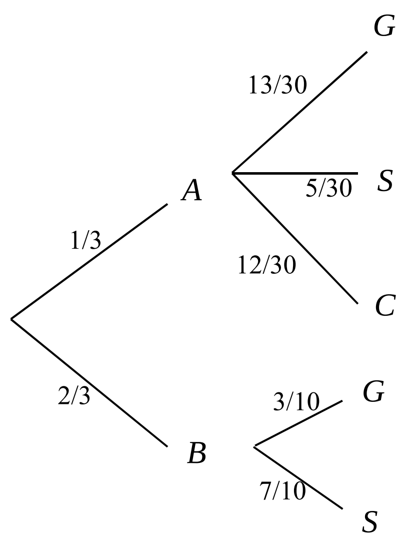
Qn	Solution
1	$k^2+kx+2x-1-x^2<0$ $x^2-(k+2)x+(1-k^2)>0$ $(k+2)^2-4(1-k^2)<0$ $k^2+4k+4-4+4k^2<0$ $5k^2+4k<0$ $k(5k+4)<0$ $-\frac{4}{5}<k<0$ 
2	$2e^x-2e^{-x}=3,$ $2e^x-\frac{2}{e^x}=3$ $2u-\frac{2}{u}=3$ $2u^2-2=3u$ $2u^2-3u-2=0$ $(2u+1)(u-2)=0$ $u=-\frac{1}{2}$ or $u=2$ $e^x=-\frac{1}{2}$ or $e^x=2$ $(\text{rej.} \because e^x>0)$ $x=\ln 2$  $x>\ln 2$

3(i)	$\dfrac{d}{dx}\left[\ln\left(3x^2-2\right)\right]=\dfrac{6x}{3x^2-2}$
3(ii)	$\displaystyle\int \frac{2x}{3x^2-2}\,dx=\frac{1}{3}\int \frac{6x}{3x^2-2}\,dx$ $\displaystyle=\frac{1}{3}\ln\left(3x^2-2\right)+C$
3(iii)	Area of region bounded = area of rectangle – area under curve $\displaystyle=\frac{2}{5}-\int_2^3\frac{2x}{3x^2-2}\,dx$ $\displaystyle=\frac{2}{5}-\frac{1}{3}\left[\ln\left(3x^2-2\right)\right]_2^3$ $\displaystyle=\frac{2}{5}-\frac{1}{3}\left(\ln 25-\ln 10\right)$ $\displaystyle=\frac{2}{5}-\frac{1}{3}\ln\left(\frac{5}{2}\right)$
4(i)	$2x+2y+\pi x=8$ $2y=8-2x-\pi x$ $y=\dfrac{8-(2+\pi)x}{2}$
4(ii)	Let A m^2 be the area of the window $A=\dfrac{1}{2}\pi x^2+2xy$ $A=\dfrac{1}{2}\pi x^2+2x\left(\dfrac{8-(2+\pi)x}{2}\right)$ $A=\dfrac{1}{2}\pi x^2+x\left[8-(2+\pi)x\right]$ $A=8x+\dfrac{1}{2}\pi x^2-\pi x^2-2x^2$ $\therefore A=\left[8x-\left(2+\dfrac{\pi}{2}\right)x^2\right]$ $=\left[8x-\left(\dfrac{4+\pi}{2}\right)x^2\right]$ m^2 (Shown)
4(iii)	$\dfrac{dA}{dx}=8-(4+\pi)x$ For stat values, $\dfrac{dA}{dx}=0$ $8-(4+\pi)x=0$ $x=\dfrac{8}{4+\pi}$

	$A=\left[8\left(\frac{8}{4+\pi}\right)-\left(\frac{4+\pi}{2}\right)\left(\frac{8}{4+\pi}\right)^2\right]$ $=\left[\frac{64}{4+\pi}-\frac{32}{(4+\pi)}\right]$ $=\frac{32}{4+\pi}\ (\text{shown})$ $\frac{d^2A}{dx^2}=-(4+\pi)<0$ A is maximum when $x=\frac{8}{4+\pi}$
5(i)	
5(ii)	Using GC, At $x=-1$, $\frac{dy}{dx}=3.1549$ (4 d.p)
5(iii)	Equation of tangent at $(-1, 2.2817)$ $y-2.2817=3.1549(x+1)$ $y=3.1549x+5.4366$
5(iv)	Coordinates of P is $(-1.7232, 0)$ (4 d.p. or 3 s.f.) Coordinates of Q is $(0, 5.4366)$ (4 d.p. or 3 s.f.) Area of triangle OPQ $=\frac{1}{2}(5.436580873)(1.723238501)$ $=4.684262737$ $=4.6843\ (4\ d.p.)$

Section B (Statistics)

6(i)	- Breakdown of 60 students into appropriate strata e.g. gender, year of study - Mention that interviewer chooses people to survey till quota is reached Disadvantage: - Sample may be biased as not students from all the years may have common lunch break. - Sample is biased as students who do not go to the canteen may not get to do the survey - Student union survey approach students (or friends) who look more friendlier or approachable in the canteen
6(ii)	Stratified sampling. **Yr 1:** $\frac{5}{12}\times 60 = 25$ **Yr 2:** $\frac{4}{12}\times 60 = 20$ **Yr 3:** $\frac{3}{12}\times 60 = 25$ Choose a random sample from each year group in size as shown.
7(i)	Ranking points 
7(ii)	$r = 0.97756 \approx 0.978$ (3 s.f.) As r is close to 1, it indicates a strong positive linear correlation between amount of time spent on revision and (x) and the ranking points obtained (y).
7 (iii)	$y = 12.356 + 0.391899985x$ $y = 12.4 + 0.392x$
7(iv)	For every additional hour spent on studying, the ranking points achieved increases by 0.392.
7(v)	For x = 170, y = 79 (nearest integer). Since the value of $r = 0.973$ is close to 1 and x = 170 is within the data range, the estimate is reliable.

8(i) (a)	Let A and B denote the event that chest A and chest B are selected respectively. Let G, S and C denote the events that a gold, silver and copper coin are chosen respectively. 
(b)	$\mathrm{P}(S)=\left(\frac{1}{3}\times\frac{5}{30}\right)+\left(\frac{2}{3}\times\frac{7}{10}\right)$ $=\frac{47}{90}$ or 0.522
(c)	$\mathrm{P}(A\mid S)=\frac{\mathrm{P}(A\cap S)}{\mathrm{P}(S)}$ $=\frac{\left(\frac{1}{3}\right)\left(\frac{5}{30}\right)}{\left(\frac{47}{90}\right)}$ $=\frac{5}{47}$ or 0.106
(ii)	$\mathrm{P}(G,G)=\left(\frac{1}{3}\times\frac{13}{30}\times\frac{12}{29}\right)+\left(\frac{2}{3}\times\frac{3}{10}\times\frac{2}{9}\right)$ $=\frac{136}{1305}$ or 0.10421 $\mathrm{P}(S,S)=\left(\frac{1}{3}\times\frac{5}{30}\times\frac{4}{29}\right)+\left(\frac{2}{3}\times\frac{7}{10}\times\frac{6}{9}\right)$ $=\frac{416}{1305}$ or 0.31877 $\mathrm{P}(C,C)=\left(\frac{1}{3}\times\frac{12}{30}\times\frac{11}{29}\right)$ $=\frac{66}{1305}$ or 0.050575

	Prob (both coins are of the same colour) $=\mathrm{P}(G,G)+\mathrm{P}(S,S)+\mathrm{P}(C,C)$ $=\frac{136}{1305}+\frac{416}{1305}+\frac{66}{1305}$ $=\frac{206}{435}$ or 0.474
9(i)	- Each interviewee is equally likely to be 'at risk' - Each interviewee is independently 'at risk' of one another
9(ii)	$\mathrm{P}(X=1)=0.00142$ $\binom{20}{1}p^{1}(1-p)^{19}=0.00142$ $20p(1-p)^{19}=0.00142$ Using GC, p = $0.36192\approx0.362$ $P(3\le X<8)=P(X\le7)-P(X\le2)$ $=0.54755\approx0.548$
9 (iii)	$P(X>4)=1-P(X\le4)$ $=0.73148$ Let Y be the no. of samples, out of 60, that have more than 4 interviewees 'at risk'. $Y\sim B(60,\ 0.73148)$ Since n = 60 large, np = 43.889>5 nq = 16.111>5 $Y\sim N(43.889,\ 11.785)$ approx $P(Y\le35)=P(Y\le35.5)$ with c.c. $=0.0072690\approx0.00727$
10(i)	Unbiased estimate of the population mean $\bar{x}=-\frac{562}{200}+960$ $=957.19$ Unbiased estimate of the population variance, s^2 $=\frac{1}{199}\left[525250-\frac{(-562)^2}{200}\right]$ $=2631.511457$ $=2631.51$
10 (ii)	$H_o:\mu=\mu_0$ $H_1:\mu\ne\mu_0$ Level of significance = 10% (= 0.10)

	Since $n=200$ is large, by Central Limit Theorem (CLT), we use a Z-test Test Statistic: $Z=\dfrac{\overline{X}-\mu_0}{s/\sqrt{n}}$ □ $\mathrm{N}(0,1)$ approximately For H_0 to not be rejected, $-1.64485<\dfrac{957.19-\mu_0}{\left(\dfrac{\sqrt{2631.511457}}{\sqrt{200}}\right)}<1.64485$ $957.19-5.96643<\mu_0<957.19+5.96643$ $951.22<\mu_0<963.16$
10 (iii)	It refers to a probability of 0.1 that the test would wrongly reject the claim that the average spending of any tourist is \$$\mu_0$, when it is actually true.
	$H_o:\mu=1100$ $H_1:\mu<1100$ Level of significance = 3% (= 0.03) Since $n=200$ is large, by Central Limit Theorem (CLT), we use a Z-test Test Statistic: $Z=\dfrac{\overline{X}-\mu}{\sigma/\sqrt{n}}$ □ $\mathrm{N}(0,1)$ approximately Using GC, with $\overline{x}=1093$, $\sigma=55$, n=200, $\mu_0=1100$ $p-\text{value}=0.0359>0.03$ Therefore we do no reject H_o and conclude that there is insignificant evidence at 3% significance level that the average spending of any tourist is less than \$1100.
11(i)	Let T and W be r.v. denoting the mass (in kg) of a randomly chosen Thai and Taiwanese guava respectively. $T\sim\mathrm{N}(0.19,0.01^2)$ and $W\sim\mathrm{N}(0.33,0.04^2)$ $\overline{T}\sim\mathrm{N}(0.19,\dfrac{0.01^2}{8})$ $\overline{T}\sim\mathrm{N}(0.19,0.0000125)$ $\mathrm{P}(\overline{T}<0.18)=0.00234$
11 (ii)	$T_1+\ldots+T_6\sim\mathrm{N}(6\times0.19,\ 6\times0.01^2)$ $W_1+\ldots+W_5\sim\mathrm{N}(5\times0.33,\ 5\times0.04^2)$ Let $D=(T_1+\ldots+T_6)-(W_1+\ldots+W_5)$

	$\mathrm{E}(D) = 6\times0.19-5\times0.33 = -0.51$ $\mathrm{Var}(D) = 6\times0.01^2+5\times0.04^2 = 0.0086$ $D \sim \mathrm{N}(-0.51,\ 0.0086)$ $\mathrm{P}(-0.5 < D < 0.5) = 0.457$
11 (iii)	Let $A = 4.20(T_1+\ldots+T_5)+2.80(W_1+\ldots+W_5)$ Let $B = 2.80(W_1+\ldots+W_{10})$ $A-B = 4.20(T_1+\ldots+T_5)+2.80(W_1+\ldots+W_5)$ $-2.80(W_1+\ldots+W_{10})$ $\mathrm{E}(A-B) = 4.20[5\times0.19]+2.80[5\times0.33]-2.80[10\times0.33]$ $= -0.63$ $\mathrm{Var}(A-B) = 4.20^2\left[5\times0.01^2\right]+2.80^2\left[5\times0.04^2\right]$ $+2.80^2\left[10\times0.04^2\right]$ $= 0.19698$ $A-B \sim \mathrm{N}(-0.63,\ 0.19698)$ $\mathrm{P}(A>B) = \mathrm{P}(A-B>0)$ $= 0.0779$